

Building Fire-Resilience Credit Model (MKM-FIRE-001)

Wolfram Oracle Benchmark: the Credit Computed Exactly

MKM Research Labs

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1 Executive Summary

This document benchmarks the Building Fire-Resilience Credit Model (MKM-FIRE-001), which prices a commercial asset's fire resilience as a containment rate, an annual loss frequency and a full-conflagration spread for the Commercial PRS Pricer. The model estimates all three by Monte Carlo. This benchmark computes them *exactly*: the per-fire jitter makes the point-of-no-return step a random variable whose law is a product of lognormal tail probabilities, and given that step the containment chain is a matrix product, so each credit is a finite sum.

The model is right, and its tables are honest. Model C reproduces its documented formulae to 2.1×10^{-16} over 90 distinct assets. The code, run at the document's own draw count and seed, reproduces 212 of the 212 figures printed in its Tables 1–12, and every printed figure lies within 2.54 standard errors of the exact credit (Table 4). Wolfram and the 50-digit reference agree to 4.3×10^{-31}.

Findings (Table 2):

1. **The production spread is a noisy draw.** Production runs 1,000 draws, and the spread is the fraction of draws that reach the point of no return. At the reach cliff the exact spread is 197.5 bp; at 1,000 draws its standard error is 44 bp, yet the pricer quotes it to 0.1 bp (Table 3). The exact credit takes about a tenth of a second per asset, so it can replace the simulation for the spread.

2. **The cliff is 197.5 bp, not 203.** The printed 203 is a sampled estimate of $10^4(1 - e^{-\lambda})$; it is within sampling error, but it is not the value.

3. **FIRE-L3, measured.** With no effective suppression the race is lost in 74% of fires, but the chain has already absorbed in all but 3.6%; the conflagration leg sees only those. The document's own limitation is confirmed and now quantified.

4. **The tall non-combustible residual does not exist.** §8 and Table 12 expect single-digit bps "in very tall non-combustible assets". The exact residual is 0.0028 bp at 20 storeys, the same as at 12: above the reach, suppression is unreachable at any height, and nothing else in the race depends on storeys.

5. **Some of the printed variation is noise.** Tables 4, 5 and 9 print containment rates from 64.0% to 68.3% across conditions, types and occupancies. Those inputs change only how often a fire starts, not how it ends: the exact containment rate is 66.8% in every row. At 20,000 draws a cell rests on a few hundred fires, and the differences are sampling error.

6. **Two statements in the document differ from the code.** Suppression becomes active at $\lceil \tau_{\text{det}} + \tau_{\text{supp}} \rceil$, not $\lceil \tau_{\text{supp}} \rceil$ (§3.3); without detection in that sum the detection level would do nothing at all. And every transition matrix carries an $S_6 \to S_7$ escalation row that the march never reads, because it stops at S_6 (§3.5).

The chain absorbs in 3.38 steps on average and intensity climbs 0.797 a step at the baseline, as §7.1 says; the non-combustible residual at 12 storeys is at most 0.053 bp, sub-bp as Table 12 says; and Table 13 is reproduced from the documented formulae.

Evidence. 6 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-27 by `python -m tools.wolfram_oracle fire001`.

2 Model Purpose and Scope

2.1 What is benchmarked

In `src/models/fire/`: Model C (`response_effectiveness.py`), the intensity race of Model B (`intensity_track.py`), the containment chain and the end-to-end `simulate_asset_fire` of Model D (`containment.py`), and the draw-level ignition of Model A (`initiation.py`), each cell built as the document's sensitivity generator built it.

2.2 What is not

Model A's modifier tables are taken as the configuration of record: the benchmark uses the code's λ_i and checks what is done with it. The initiation class, which the document notes is cosmetic, does not enter the credit. Every parameter is an engineering-judgement seed (FIRE-L1); this benchmark shows the model computes what it documents, not that the seeds are right. It addresses validation question Q8, not independent validation (Q9).

3 Mathematical Framework

The race. Before suppression bites, intensity is $\min(t\,g', C')$, with $g' = g\,e^{\sigma_g Z_1}$ and $C' = I_{\max} e^{\sigma_c Z_3}$, and suppression bites at $\tau_{\det} + \tau'$, $\tau' = \tau_{\text{supp}} e^{\sigma_\tau Z_2}$, the Z_i independent standard normals. The point of no return latches at the first t with $t\,g' > I_{\text{crit}}$, provided $C' > I_{\text{crit}}$ and $t < \tau_{\det} + \tau'$ (at $t = 1$ for a combustible frame suppression cannot reach). So

$$P(K = t) = P\Big(\tfrac{I_{\text{crit}}}{t} < g' \le \tfrac{I_{\text{crit}}}{t-1}\Big)\, P(\tau' > t - \tau_{\det})\, P(C' > I_{\text{crit}})\,, \tag{1}$$

each factor a lognormal tail.

The chain. With $M = wM_{AA} + (1-w)M_{\text{weak}}$ before the latch and M_{PNR} from it, and the terminal states made absorbing,

$$\pi = \sum_{t=1}^{T} P(K = t)\; e_0 M^{t-1} M_{\text{PNR}}^{T-t+1} + P(K > T)\; e_0 M^T, \tag{2}$$

with $T = 200$ the step budget; a point of no return is recorded only if the chain is still transient after $t-1$ steps. The credit is the containment rate π_{S_5}, the loss frequency $q(\pi_{S_6} + \pi_{S_7})$ and the spread $10^4\, q\, P(\text{PNR recorded})$, with $q = 1 - e^{-\lambda h}$ the probability a draw ignites.

4 Benchmark Method

Model C and the two sums above are evaluated with `mpmath` at 50 digits, from the document rather than the code; lognormal tails are taken from `erfc`, so a 10^{-140} probability keeps its digits. Wolfram evaluates the same sums independently — the race law from `Erfc` survival functions, the chain with `FoldList` and `MatrixPower` — for each of the 90 distinct assets behind the 108 printed cells. The model runs each cell at 20,000 draws with seed 20260606, as the generator did, and each of its figures is judged in standard errors of the exact value; the containment rate's on the expected number of fires.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Model C: response profile against the documented formulae (s3.2)	2.1×10^{-16}	1.0×10^{-12}	pass
Containment rate: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	2.52	4	pass
Loss frequency: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.78	4	pass
Fire spread: model's Monte Carlo against the exact credit, worst $\lvert z \rvert$	1.59	4	pass
Fires unresolved at the 200-step budget	3.4×10^{-17}	1.0×10^{-12}	pass
Wolfram agrees with the exact reference (outcome probabilities, q)	4.3×10^{-31}	1.0×10^{-12}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§7, Tables 1–12	the tables come from the running model	the code reproduces 212 of 212 printed figures at the generator's seed	met
§7, Tables 1–12	the figures are the model's credit	every printed figure within 2.54 standard errors of the exact value	met
§7.1	the chain absorbs in ≈ 3 steps	3.38 steps (baseline)	met
§7.1	intensity climbs ≈ 0.8 a step	0.7969 (baseline)	met
Table 12	non-combustible residual is sub-bp	at most 0.053 bp	met
Table 12, §8	single-digit bps in very tall non-combustible assets	0.0028 bp at 20 storeys: height does not enter an unreachable race	**not met**
Table 2	the reach cliff is 203 bp	exactly 197.5 bp; 203 is a draw ($z = 0.56$)	met
Table 13	construction levers	28 of 28 reproduced from the formulae	met
FIRE-L3	the chain absorbs before the race is lost	no suppression: race lost in 74.3% of fires, recorded in 3.56%	finding
§3.5	S_6 is absorbing	the march stops at S_6, but every matrix carries an unused $S_6 \to S_7$ row (0.15, 0.5, 0.65)	finding
§3.3	suppression is active from step $\lceil \tau_{\text{supp}} \rceil$	the code, and the config's own definition, use $\lceil \tau_{\text{det}} + \tau_{\text{supp}} \rceil$	finding
Tables 4, 5, 9	containment 64.0–68.3% by condition, type, occupancy	exactly 66.8% for all: these inputs move only the ignition rate	finding

5.2 The spread at the production draw count

Table 3: The fire spread and its sampling error at the production draw count (1,000)

Cell	Exact (bp)	Model at 20k (bp)	Std error at production (bp)
T1/Not assessed	7.40	8.5	8.6
T2/12 storeys	197.52	203.0	44.0
T3/Partial	6.77	5.0	8.2
T12/Timber frame	207.81	216.0	45.1

5.3 Every printed cell

Table 4: Every printed cell against the exact credit: containment (%), loss frequency (%), fire spread (bp); $|z|$ is the worst printed figure in standard errors at 20,000 draws

		Printed			Exact			$\lvert z \rvert$
T	Cell							
1	Not assessed	47.0	1.150	8.5	45.6	1.131	7.4	0.59
1	Partial	63.1	0.750	0.0	56.8	0.854	0.3	2.54
1	Meets minimum	66.7	0.675	0.0	66.8	0.657	0.0	0.32
1	Enhanced	78.1	0.445	0.0	77.1	0.452	0.0	0.45
1	Verified	86.8	0.260	0.0	87.9	0.227	0.0	1.00
2	1 storeys	63.1	0.750	0.0	56.8	0.854	0.3	2.54
2	3 storeys	63.1	0.750	0.0	56.8	0.854	0.3	2.54
2	4 storeys	63.1	0.750	0.0	56.8	0.854	0.3	2.54

Table 4: Every printed cell against the exact credit: containment (%), loss frequency (%), fire spread (bp); $|z|$ is the worst printed figure in standard errors at 20,000 draws

T	Cell	Printed			Exact			$\lvert z\rvert$
2	5 storeys	0.0	2.030	203.0	0.0	1.975	197.5	0.56
2	8 storeys	0.0	2.030	203.0	0.0	1.975	197.5	0.56
2	12 storeys	0.0	2.030	203.0	0.0	1.975	197.5	0.56
2	20 storeys	0.0	2.030	203.0	0.0	1.975	197.5	0.56
3	Not assessed	63.5	0.740	4.5	62.0	0.751	7.6	1.59
3	Partial	63.5	0.740	5.0	63.3	0.725	6.8	0.96
3	Meets minimum	66.7	0.675	0.0	66.8	0.657	0.0	0.32
3	Enhanced	66.0	0.690	0.0	67.8	0.636	0.0	0.96
3	Verified	66.3	0.685	0.0	68.8	0.615	0.0	1.26
4	Excellent	68.3	0.525	0.0	66.8	0.532	0.0	0.58
4	Good	67.0	0.625	0.0	66.8	0.595	0.0	0.56
4	Fair	66.7	0.675	0.0	66.8	0.657	0.0	0.32
4	Poor	66.5	0.790	0.0	66.8	0.749	0.0	0.67
4	Very poor	64.6	0.970	0.0	66.8	0.872	0.0	1.48
5	Office	66.7	0.675	0.0	66.8	0.657	0.0	0.32
5	Retail	64.6	1.190	0.0	66.8	1.076	0.0	1.56
5	Hotel	66.8	1.430	0.0	66.8	1.419	0.0	0.15
5	Leisure	66.4	1.250	0.0	66.8	1.220	0.0	0.38
5	Healthcare	65.4	0.975	0.0	66.8	0.899	0.0	1.13
5	MultiFamily	64.0	1.160	0.0	66.8	1.028	0.0	1.85
5	MixedUse	66.4	1.520	0.0	66.8	1.491	0.0	0.34
5	Other	66.4	1.520	0.0	66.8	1.491	0.0	0.34
6	3 / Partial	–	–	0.0	56.8	0.854	0.3	0.81
6	3 / Meets minimum	–	–	0.0	66.8	0.657	0.0	0.10
6	3 / Enhanced	–	–	0.0	77.1	0.452	0.0	0.00
6	3 / Verified	–	–	0.0	87.9	0.227	0.0	0.00
6	5 / Partial	–	–	203.0	0.0	1.975	197.5	0.56
6	5 / Meets minimum	–	–	0.0	66.8	0.657	0.0	0.10
6	5 / Enhanced	–	–	0.0	77.1	0.452	0.0	0.00
6	5 / Verified	–	–	0.0	87.9	0.227	0.0	0.00
6	8 / Partial	–	–	203.0	0.0	1.975	197.5	0.56
6	8 / Meets minimum	–	–	0.0	66.8	0.657	0.0	0.11
6	8 / Enhanced	–	–	0.0	77.1	0.452	0.0	0.00
6	8 / Verified	–	–	0.0	87.9	0.227	0.0	0.00
6	20 / Partial	–	–	203.0	0.0	1.975	197.5	0.56
6	20 / Meets minimum	–	–	0.0	66.8	0.657	0.0	0.11
6	20 / Enhanced	–	–	0.0	77.1	0.452	0.0	0.00
6	20 / Verified	–	–	0.0	87.9	0.227	0.0	0.00
7	Reinforced concrete	87.7	0.250	0.0	87.9	0.238	0.0	0.34
7	Concrete frame	83.5	0.335	0.0	82.9	0.338	0.0	0.33
7	Steel frame	78.8	0.430	0.0	77.9	0.437	0.0	0.44
7	Brick and block	78.8	0.430	0.0	77.9	0.437	0.0	0.44
7	Modern methods	63.8	0.735	0.0	63.5	0.722	0.0	0.22
7	Mixed construction	0.0	2.030	203.0	0.0	1.975	197.5	0.56
7	Timber frame	0.0	2.030	203.0	0.0	1.975	197.5	0.56
8	Not assessed	66.9	0.715	0.0	66.8	0.691	0.0	0.41
8	Partial	66.7	0.675	0.0	66.8	0.657	0.0	0.32
8	Meets minimum	66.7	0.675	0.0	66.8	0.657	0.0	0.32

Table 4: Every printed cell against the exact credit: containment (%), loss frequency (%), fire spread (bp); $|z|$ is the worst printed figure in standard errors at 20,000 draws

T	Cell	Printed			Exact			$\lvert z\rvert$
8	Enhanced	66.7	0.675	0.0	66.8	0.657	0.0	0.32
8	Verified	67.3	0.645	0.0	66.8	0.622	0.0	0.41
9	Fully occupied	66.7	0.675	0.0	66.8	0.657	0.0	0.32
9	Partially vacant	67.3	0.615	0.0	66.8	0.591	0.0	0.43
9	Vacant	67.4	0.575	0.0	66.8	0.559	0.0	0.31
10	Reinforced concrete / 3	–	–	0.0	87.9	0.238	0.0	0.00
10	Reinforced concrete / 5	–	–	0.0	87.9	0.238	0.0	0.00
10	Reinforced concrete / 8	–	–	0.0	87.9	0.238	0.0	0.00
10	Reinforced concrete / 20	–	–	0.0	87.9	0.238	0.0	0.00
10	Steel frame / 3	–	–	0.0	77.9	0.437	0.0	0.00
10	Steel frame / 5	–	–	0.0	77.9	0.437	0.0	0.00
10	Steel frame / 8	–	–	0.0	77.9	0.437	0.0	0.00
10	Steel frame / 20	–	–	0.0	77.9	0.437	0.0	0.00
10	Modern methods / 3	–	–	0.0	63.5	0.722	0.0	0.00
10	Modern methods / 5	–	–	0.0	63.5	0.722	0.0	0.00
10	Modern methods / 8	–	–	0.0	63.5	0.722	0.0	0.00
10	Modern methods / 20	–	–	0.0	63.5	0.722	0.0	0.00
10	Mixed construction / 3	–	–	0.0	56.8	0.854	0.3	0.81
10	Mixed construction / 5	–	–	203.0	0.0	1.975	197.5	0.56
10	Mixed construction / 8	–	–	203.0	0.0	1.975	197.5	0.56
10	Mixed construction / 20	–	–	203.0	0.0	1.975	197.5	0.56
10	Timber frame / 3	–	–	3.5	55.6	0.877	4.0	0.35
10	Timber frame / 5	–	–	203.0	0.0	1.975	197.5	0.56
10	Timber frame / 8	–	–	203.0	0.0	1.975	197.5	0.56
10	Timber frame / 20	–	–	203.0	0.0	1.975	197.5	0.56
11	Reinforced concrete / Partial	87.7	–	–	87.9	0.238	0.0	0.15
11	Reinforced concrete / Meets minimum	87.7	–	–	87.9	0.238	0.0	0.15
11	Reinforced concrete / Enhanced	87.7	–	–	87.9	0.238	0.0	0.15
11	Reinforced concrete / Verified	87.1	–	–	87.9	0.226	0.0	0.50
11	Steel frame / Partial	78.8	–	–	77.9	0.437	0.0	0.44
11	Steel frame / Meets minimum	78.8	–	–	77.9	0.437	0.0	0.44
11	Steel frame / Enhanced	78.8	–	–	77.9	0.437	0.0	0.44
11	Steel frame / Verified	86.8	–	–	87.9	0.227	0.0	0.65
11	Modern methods / Partial	63.8	–	–	63.5	0.722	0.0	0.14
11	Modern methods / Meets minimum	66.7	–	–	66.8	0.656	0.0	0.03
11	Modern methods / Enhanced	78.1	–	–	77.1	0.452	0.0	0.45
11	Modern methods / Verified	86.8	–	–	87.9	0.227	0.0	0.65
11	Mixed construction / Partial	0.0	–	–	0.0	1.975	197.5	0.00
11	Mixed construction / Meets minimum	66.7	–	–	66.8	0.657	0.0	0.03
11	Mixed construction / Enhanced	78.1	–	–	77.1	0.452	0.0	0.45
11	Mixed construction / Verified	86.8	–	–	87.9	0.227	0.0	0.65
11	Timber frame / Partial	0.0	–	–	0.0	1.975	197.5	0.00
11	Timber frame / Meets minimum	66.5	–	–	66.7	0.658	0.2	0.08
11	Timber frame / Enhanced	78.1	–	–	77.1	0.452	0.0	0.45
11	Timber frame / Verified	86.8	–	–	87.9	0.227	0.0	0.65
12	Reinforced concrete	87.5	0.270	0.0	87.9	0.251	0.0	0.55
12	Concrete frame	83.3	0.360	0.0	82.9	0.356	0.0	0.23
12	Steel frame	78.7	0.460	0.0	77.9	0.460	0.0	0.40

Table 4: Every printed cell against the exact credit: containment (%), loss frequency (%), fire spread (bp); $|z|$ is the worst printed figure in standard errors at 20,000 draws

T	Cell	Printed			Exact			$\lvert z\rvert$
12	Brick and block	78.7	0.460	0.0	77.9	0.460	0.0	0.40
12	Modern methods	63.9	0.780	0.0	63.4	0.760	0.1	0.33
12	Mixed construction	0.0	2.160	216.0	0.0	2.078	207.8	0.81
12	Timber frame	0.0	2.160	216.0	0.0	2.078	207.8	0.81

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The credit is shown to be computed as documented; every parameter remains an uncalibrated seed (FIRE-R1, R2).

2. **Spread precision.** Compute the spread exactly, or raise the draw count until its standard error is below the precision quoted; at 1,000 draws it is a draw, not a price.

3. **Documentation.** §3.3 should put detection into the activation step; §3.5 should say the S_6 rows are unused, or the march should use them; Table 2 should quote the cliff as $10^4(1 - e^{-\lambda})$; §8's tall-residual sentence should be removed. At a combustibility exactly on the threshold a frame is both combustible and given the low fuel cap; no configured type sits there.

7 Document History

Date	Version	Description	Author
27-Oct-2026	0.1	First issue	David K Kelly